

Research advance: Unexpected plasticity in the life cycle of *Trypanosoma brucei*

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eLife Assessment

The manuscript concerns a **fundamental** and controversial question in *Trypanosoma brucei* biology and the parasite life cycle, providing further evidence that slender bloodstream forms can indeed infect Tsetse flies. The study is **solid** in design and execution, and addresses several criticisms made of the authors' earlier work. Nevertheless, some of the main conclusions are only partially supported: one issue is how, precisely, a "slender" bloodstream form is defined, and discrepancies with some results from other laboratories remain unexplained.

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Abstract

We have previously shown that the slender form of *Trypanosoma (T.) brucei* is able to infect teneral tsetse flies, develop to the first fly form, which is the procyclic form, and complete the life cycle in the insect vector (Schuster et al., 2021). Further, analysis of the transmission index (TI; defined as the number of salivary gland infections relative to the number of midgut infections) revealed a higher TI for slender as compared to stumpy forms under laboratory conditions, which included the addition of *N*-acetyl-glucosamine (NAG) to the infective bloodmeal.

These findings challenge the prevailing view of the life cycle, according to which only stumpy forms are considered infective to tsetse flies.

Here, we show that slender trypanosomes can infect both male and female tsetse flies, irrespective of their teneral status, in the absence of supplements in the bloodmeal.

Additionally, an RNA-sequencing time course was performed on both slender and stumpy cells during their transition into procyclic forms. This analysis revealed that slender and stumpy form trypanosomes remain transcriptionally distinct throughout differentiation into the procyclic form. Furthermore, while the protein associated with differentiation 1 (PAD1) remains essential for the transition, slender cells do not require expression of other hallmark

stumpy form traits, such as cell cycle arrest or the shortening of their flagella or microtubule corset. Instead, slender trypanosomes are able to transition directly into procyclic forms.

Taken together, these findings demonstrate that while slender cells of *T. brucei* follow distinct routes to become the procyclic form, they are capable of infecting both teneral and non-teneral tsetse flies, thereby contributing to the transmission and spread of these African parasites.

Introduction

In the bloodstream of a mammalian host, two main forms of *Trypanosoma (T.) brucei* can be observed, the long slender and the short stumpy bloodstream form (bsf).

The slender cell is proliferative whereas the stumpy cell has undergone cell-cycle arrest in G1/G0 phase. During division, slender cells release peptidases, most significantly oligopeptidase B and metallocarboxypeptidase 1 (Tetty et al., 2022 [↗](#)), generating a pool of oligopeptides in the blood and tissues around them. These essential peptides, together with possible other unknown components, are part of the molecular cocktail collectively called the Stumpy Induction Factor (SIF) (Vassella et al., 1997 [↗](#); Reuner et al., 1997 [↗](#); Bossard et al., 2013 [↗](#); Moss et al., 2015 [↗](#); Rojas et al., 2019 [↗](#)). After exceeding a certain threshold, SIF causes the proliferative slender cells to change into the cell-cycle arrested stumpy forms (Vassella et al., 1997 [↗](#); Reuner et al., 1997 [↗](#)). This transition is accompanied by shortening of the flagellum, cell-cycle arrest and other molecular changes, such as the remodelling of the mitochondrion to a cristate structure and expression of Krebs cycle enzymes. The trypanosomes also begin to express the Protein Associated with Differentiation 1 (PAD1), a member of the carboxylate-transporter protein family, on their surface. PAD1 functions as a transducer of the signal that triggers differentiation into the procyclic form, the first fly form that develops in the tsetse midgut (Reuner et al., 1997 [↗](#); Dean et al., 2009 [↗](#)).

Cell-cycle arrest is lethal for stumpy forms, as they die, if they are not taken up by the tsetse fly within a few days. Thus, their purpose in the life cycle of *T. brucei* has long been debated, with the two most accepted theories being:

1. Stumpy forms are needed for the regulation of the parasitaemia in the host.
2. Due to pre-adaptation for life in the insect vector, they are the only form able to infect the tsetse fly (Robertson, 1912 [↗](#); Wijers & Willett, 1960 [↗](#); Fenn & Matthews, 2007 [↗](#); MacGregor et al., 2012 [↗](#); Matthews et al., 2015 [↗](#); Silvester et al., 2017 [↗](#)).

In 2021, our laboratory reported unexpected plasticity in the life cycle of *T. brucei*. We found that even a single slender cell can infect the tsetse fly without exhibiting morphological and biochemical manifestations that define a stumpy cell (Schuster et al., 2021 [↗](#)).

Instead, slender cells turn on the essential PAD1 pathway (Dean et al., 2009 [↗](#)) and transit directly into procyclic forms - all whilst continuously dividing. We suggested that the ability of slender bloodstream forms to infect the tsetse fly vector would, at least in part, solve the transmission paradox, which refers to the low blood parasitaemia observed in chronically infected hosts and the small bloodmeal volume of tsetse flies, both of which make it unlikely that a stumpy form would be ingested (Capewell et al., 2019 [↗](#)).

However, our original study was criticised (Matthews & Larcombe, 2022 [↗](#); Ngoune et al., 2025 [↗](#)) for mainly four reasons:

1. As routinely done in tsetse laboratories, we supplemented all infectious bloodmeals with the immune-suppressive chemical *N*-acetyl-glucosamine (NAG), which is known to enhance tsetse midgut infections for stumpy forms but has no effect on subsequent salivary gland infections (Peacock et al., 2012 [↗](#)). It was argued that this treatment allowed slender forms to infect the fly.
2. We used teneral flies (which have not yet taken a bloodmeal and younger than 3 days) for all infection experiments. This led to the question if the slender trypanosomes might just be able to infect young flies.
3. The use of male flies is common practice for studying trypanosome infections. Nevertheless, could it be possible that slender trypanosomes can infect male but not female tsetse flies?
4. We had shown that slender trypanosomes express the differentiation marker *pad1* while becoming procyclic. This was taken as a proof that slender cells must turn into stumpy cells before becoming procyclic.

To address the critique further than already done (Lisack et al., 2022 [↗](#)), we have systematically conducted additional experiments. We found that slender form trypanosomes can indeed infect both teneral and non-teneral flies (i. e. flies that have taken at least one non-infectious bloodmeal and are more than 72 hours post eclosion (hpe)), even in the absence of immune suppressing chemicals.

Using an RNA-sequencing time course, we confirmed that slender forms are able to transition directly to procyclic forms without becoming stumpy forms.

Taken together these new findings further highlight the plasticity in the life cycle of *T. brucei* as well as the ability of slender trypanosomes to contribute to the spread of these parasites to new hosts.

Results

It has been shown that *N*-acetyl-glucosamine (NAG) aids stumpy trypanosome infection of the fly midgut, while not influencing the number of subsequent salivary gland infections (Peacock et al., 2006 [↗](#), 2012 [↗](#)).

To exclude the possibility that use of the lectin-inhibitor NAG rendered the tsetse fly midgut artificially permissive to slender trypanosome infections, tsetse flies were infected with slender cells in either untreated blood or NAG-supplemented blood.

As described in Schuster *et al.*, a stumpy marker cell line (NLS-GFP:PAD1 3'UTR) was used to confirm that less than 1% PAD1-positive cells were present in any slender culture.

Infection rates in the midgut and proventriculus showed negligible differences between non-supplemented and NAG-supplemented infections (**Figure 1A** [↗](#) and **Supplementary Figure 1** [↗](#)). Midgut infections reached 11.6% with and 9.2% without NAG (**Figure 1A** [↗](#)) while proventriculus infections occurred in 9.4% and 8.3%, respectively (**Figure 1A** [↗](#)). Salivary gland infections were observed in 3.6% of flies infected with and 0.9% without NAG, however, this difference was not statistically significant ($p > 0.05$) (**Figure 1A** [↗](#)).

The use of male flies is common practice for studying tsetse infections, as they show higher salivary gland infection rates than female flies (Jackson, 1949 [↗](#); Maudlin et al., 1990 [↗](#); Maudlin, 1991 [↗](#)). However, because female flies have a longer lifespan, they cannot be disregarded as

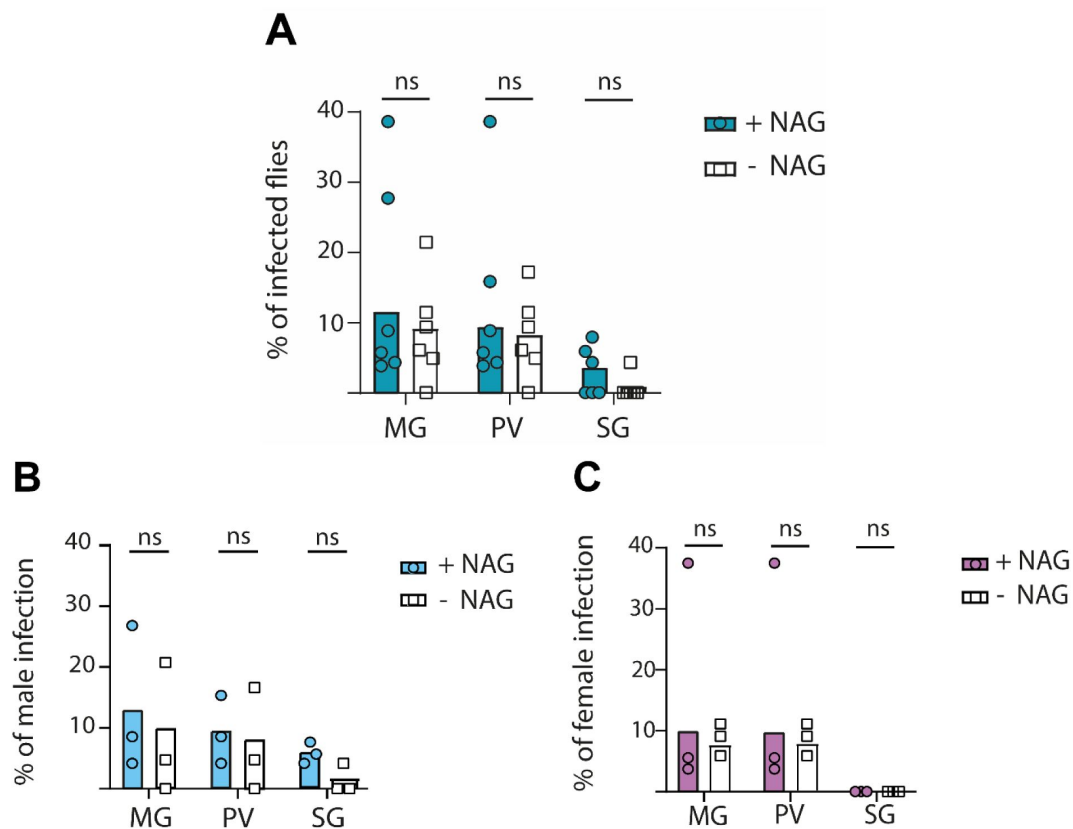


Figure 1

Infection rates (%) of slender *T. brucei* in tsetse flies, with and without NAG supplementation.

Flies of both genders were fed 4 slender cells per bloodmeal, with or 60 mM *N*-acetyl-glucosamine (NAG). Slender cells were harvested and checked for PAD1 signal to confirm slender identity before infection (< 1% PAD1 positive). Infections were performed in triplicates with approximately 20 flies/replicate. The midgut (MG), proventriculus (PV) and salivary glands (SG) of all flies were dissected after 35 days to check for infection. Bar graphs show mean infection rates across replicates, with individual dots representing infection percentages per replicate. Fisher's exact test was used on the mean infection rates to determine significance. ns = not significant ($p > 0.05$). Infection rates are shown for both sexes (A), male (B), and female flies (C).

potential important contributors to parasite transmission in the wild (Maudlin et al., 1990 [↗](#)). Therefore, the influence of fly sex on infection rates was assessed by using the same dataset as in **Figure 1A** [↗](#) (**Figure 1B and C** [↗](#)). Using NAG-supplemented blood, 8.0% of male flies exhibited midgut, 5.8% proventriculus and 3.6% salivary gland infections. In the absence of NAG, midgut, proventriculus, and salivary gland infections were observed in 5.5%, 4.6%, and 0.9% of male flies, respectively (**Figure 1B** [↗](#)). Fisher's exact test revealed no significant differences in infection rates between NAG-supplemented and non-supplemented groups for male flies (**Figure 1B** [↗](#)).

Among female flies, 3.6% harboured midgut and 3.7% proventriculus infections, regardless of NAG supplementation (**Figure 1C** [↗](#)). No salivary gland infections were identified in female flies (**Figure 1C** [↗](#)). The total percentage of fly infections and their corresponding replicates are provided in **Supplementary Figure 1** [↗](#).

In conclusion, NAG has a negligible effect on slender infections in tsetse flies and even low numbers of slender cells can infect flies without the aid of immune suppressing compounds.

Although infection rates for teneral flies are relatively low, commonly ranging between 10-30%, they are more susceptible to trypanosome infection than non-teneral flies (Peacock et al., 2012 [↗](#)). Consequently, the use of teneral flies is standard practice in *T. brucei* infection experiments, and we have exclusively used them in our previous infections. It is possible, however, that in nature, older flies contribute more to the spread of trypanosomes than younger flies. As such, it was of interest to test if slender trypanosomes could also infect non-teneral flies (Van Hoof L. et al., 1937 [↗](#); Wijers, 1958 [↗](#); Otieno et al., 1983 [↗](#); Walshe et al., 2011 [↗](#); Matthews & Larcombe, 2022 [↗](#)). Therefore, to ensure our flies would survive for at least 30 days to allow infection assessment, we defined our non-teneral flies as those infected between 144 and 168 hours post eclosion (hpe). These flies were fed two non-infectious bloodmeals prior to the third infectious bloodmeal, all spaced at least two days apart.

For these experiments, we again used the cell line containing the NLS-GFP:PAD1 3'UTR stumpy reporter and a cytoplasmic red fluorescent protein (td Tomato) (Reuter et al., 2023 [↗](#)). To ensure pure populations (Schuster et al., 2021 [↗](#)), fluorescence activated cell sorting (FACS) was performed to separate PAD1-positive (stumpy) from PAD1-negative (slender) cells. To confirm sorting success, populations were subsequently examined by fluorescence microscopy for PAD1 signal. Stumpy cells were also sorted to remove any slender cells and to keep conditions constant.

Additionally, the growth of sorted slender cells was monitored *in vitro* to ensure that sorting did not affect parasite fitness by causing any stress (Quintana et al., 2021 [↗](#)) (**Supplementary Figures 3** [↗](#) and 4). Only then did we proceed with infections, using either sorted slender or sorted stumpy cells to infect male and female, teneral or non-teneral flies. With the confidence of having pure slender or stumpy populations, we increased parasite concentrations to 1×10^6 cells/ml for infection. Flies were dissected 30 days post infection and their organs examined for parasite presence.

Infection data show that slender trypanosomes were able to establish infections in non-teneral flies at rates comparable to stumpy forms (**Figure 2A** [↗](#) and **Supplementary Figure 2** [↗](#)). Midgut infections were detected in 6.4% of flies infected with slender and 6.6% of flies infected with stumpy cells. Similarly, 5.1% and 4.4% of flies exhibited proventriculus infections, and 3.8% and 1.5% salivary gland infections, respectively. Statistical analysis using Fisher's exact test revealed no significant difference for any fly organ in slender or stumpy infections.

Consistent with published data on stumpy infections (Peacock et al., 2012 [↗](#)), male flies showed the highest infection rates (**Figure 2B** [↗](#)). Importantly, we found that non-teneral female flies can also be infected with slender cells. Midgut infections were observed in 9.5% of non-teneral female flies infected with slender cells and 5.6% with stumpy cells (**Figure 2C** [↗](#)). Although overall

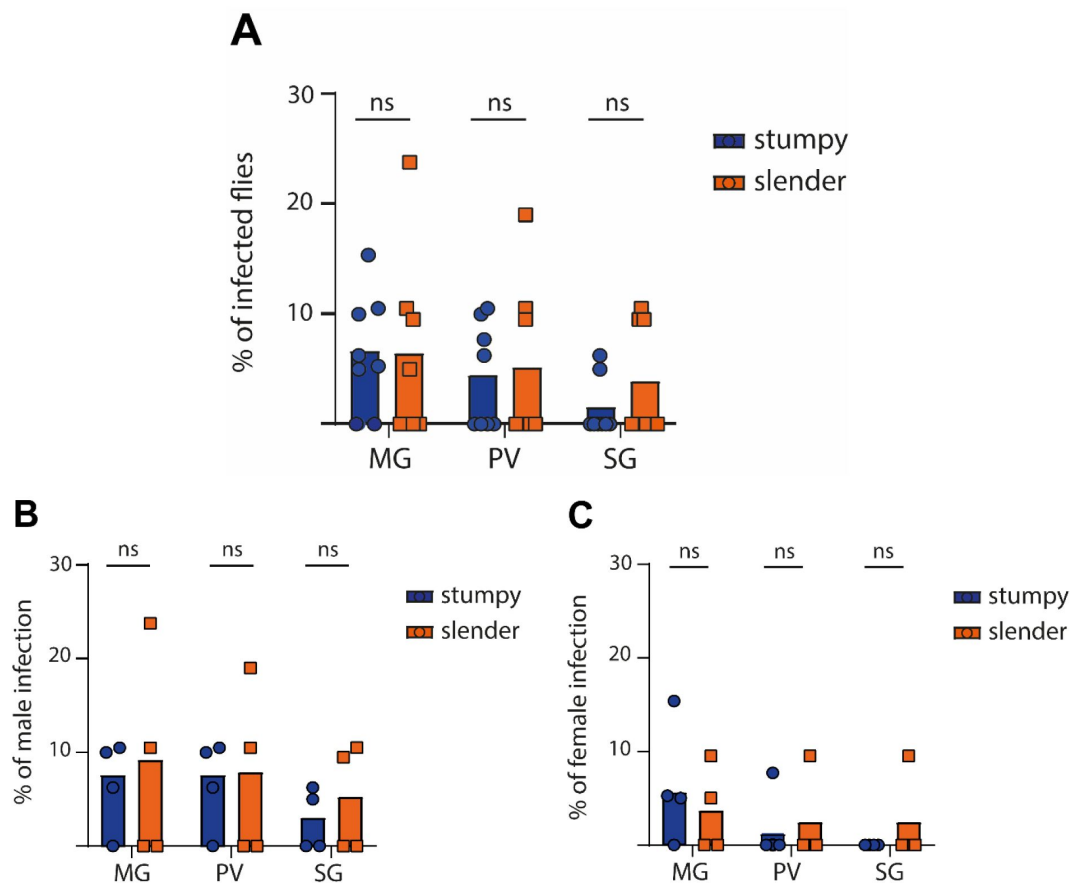


Figure 2

Infection rates (%) of slender and stumpy *T. brucei* in non-teneral tsetse flies.

Non-teneral flies were infected between 144 and 168 hpe with either slender (orange) or stumpy (blue) parasites, after receiving two non-infectious bloodmeals beforehand. The tdTomato NLS-GFP:PAD1 3'UTR cell line enabled FACS to separate stumpy (PAD1 positive, GFP in nucleus) and slender (PAD1 negative, no nuclear fluorescence) prior to infection. Infections were performed in quadruplets with roughly 20 flies/replicate. Midgut (MG), proventriculus (PV), and salivary glands (SG) of all flies were dissected after 30-35 days to check for parasite presence. Bar graphs show mean infection rates across replicates, with individual dots representing infection percentages per replicate. Fisher's exact test was used on the mean infection rates to determine significance; ns = not significant ($p > 0.05$). Infection rates are presented for both sexes (A), males (B), and females (C).

numbers were low, it is noteworthy that only slender cells caused any salivary gland infections in non-teneral female flies (**Figure 2C**). A table of all percentages of fly infections, replicates, and sex can be found in **Supplementary Figure 2**.

We have previously shown that slender and stumpy forms both differentiate to procyclic forms with comparable kinetics (Schuster et al. 2021). To assess whether slender cells transcriptionally transition through a stumpy state or differentiate directly into the procyclic form, we performed RNA sequencing of slender and stumpy parasites throughout differentiation. For this, we used the same cell line as in Schuster et al. (2021), containing an EP1:YFP fusion protein and NLS-GFP:PAD1 3'UTR stumpy reporter. *In vitro* differentiation of both slender and stumpy cultures was induced by glucose depletion, addition of *cis*-aconitate, and temperature drop to 27°C (Mowatt & Clayton, 1987; Richardson et al., 1988; Roditi et al., 1989; Ziegelbauer et al., 1990; Matthews & Gull, 1994; Dean et al., 2009).

Slender or stumpy cells were collected in triplicates at five time-points throughout differentiation, 0, 8, 15, 24, and 72 hours (hrs), and subjected to paired-end sequencing (**Figure 3**). Hierarchical clustering identified two biological replicates as outliers, which were excluded from further analysis.

Principal component analysis (PCA) was performed, with the 1st principal component accounting for 32.2% and the 2nd principal component for 11.1% of total variance (**Figure 3A**). When comparing corresponding timepoints between slender and stumpy forms, for example, slender 8 hrs versus stumpy 8 hrs, both cell types initially exhibit a high number of differentially expressed genes. This number gradually decreases, reaching zero at 72 hrs (**Figure 3B**; **Supplementary Figure 5**). Interestingly, when comparing all time points, an additional convergence with only 22 differentially expressed genes is observed between slender 15 hrs and stumpy 0 hrs (**Figure 3C**; **Supplementary Figure 6**).

At first glance, this might suggest that slender cells transition into the stumpy form at this point before proceeding to differentiate into procyclic forms in the same way as stumpy cells. However, comparison of subsequent timepoints reveals a different pattern: at slender 24 hrs and stumpy 8 hrs, the number of differentially expressed genes increases markedly to 316. This elevated number remains relatively consistent until both cell types converge at 72 hrs, where no differentially expressed genes are detected (**Figure 3B**; **Supplementary Figures 5** and 6).

In line with these findings, Gene Ontology (GO) enrichment analysis reveals that gene expression profiles of slender and stumpy cells throughout differentiation are associated with distinct molecular functions and biological processes. In slender cells, the most significant GO terms are related to extracellular structure and matrix organization, glycolytic processes and proteolysis (**Supplementary Figure 7**). In contrast, stumpy cells show enrichment for genes that are involved in RNA processing, ribosome biogenesis as well as cellular component biogenesis, consistent with them re-entering the cell cycle to become procyclic forms (**Supplementary Figure 7**). Genes associated with the procyclic form, like the procyclin surface proteins EP1 and EP2, as well as pyruvate phosphate dikinase (PPDK), start with low expression levels for both, slender and stumpy forms. However, expression increases to procyclic levels by only 8 hrs in stumpy cells but more gradually in slender cells, reaching comparable levels only by 72 hrs (**Supplementary Figure 8**).

Collectively, these data indicate that differentiation in slender and stumpy trypanosomes proceeds via distinct gene expression programs. Rather than following a shared trajectory, each form activates a unique set of genes to transition into the first fly form, the procyclic form (**Figure 4**).

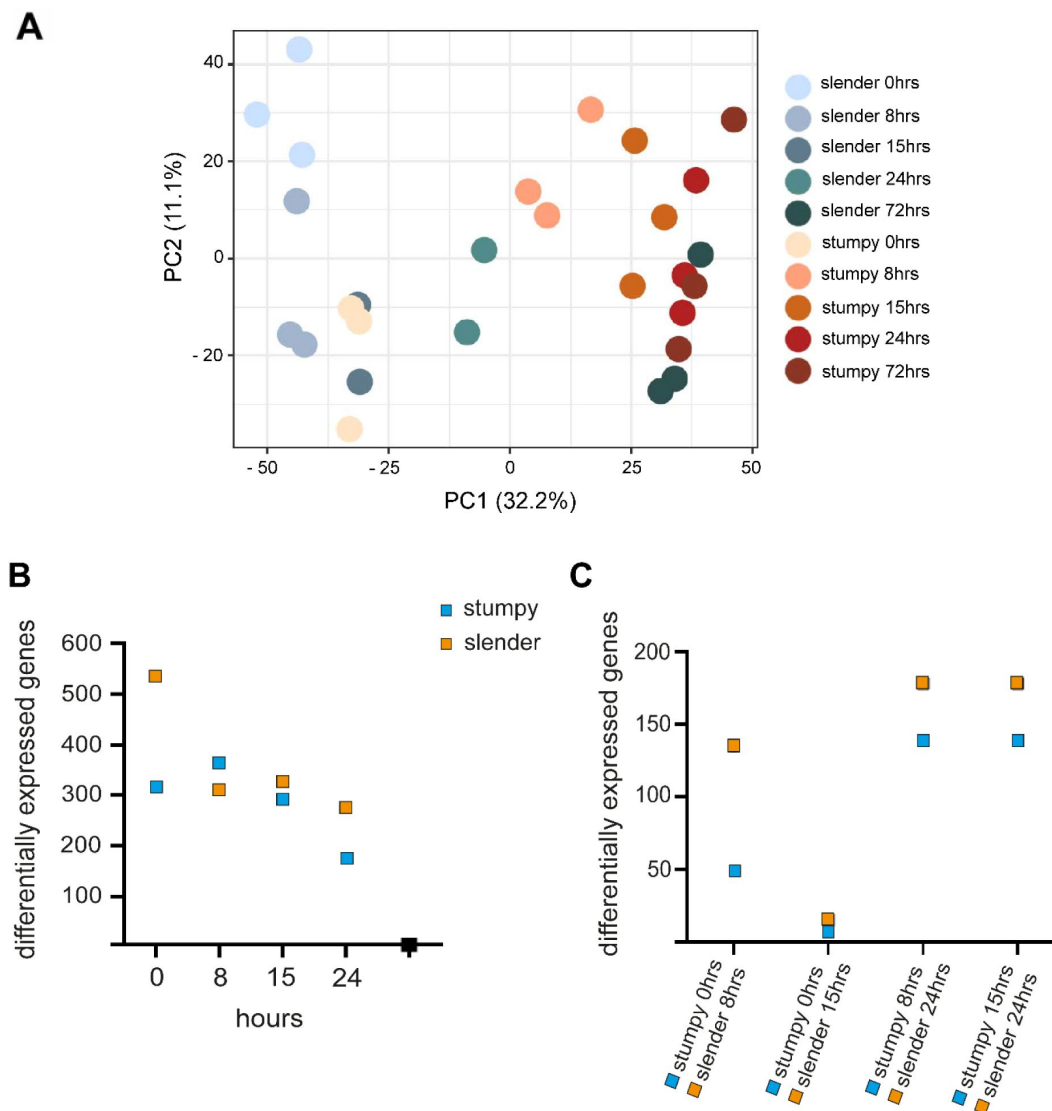


Figure 3

RNA sequencing of slender and stumpy trypanosomes differentiating into procyclic forms.

A: Principal Component Analysis (PCA) showing the transcriptional progression to procyclic forms for slender (blue/green) and stumpy (orange/red) cells. The trajectories remain distinct until converging at 72hrs. B: Number of differentially expressed genes between slender and stumpy forms at corresponding time points during differentiation. Genes with an absolute $\log_2FC > 2$ and an adjusted p -value < 0.01 were classified as differentially expressed. Corresponding volcano plots and detailed gene counts can be found in [Supplementary Figure 5](#). C: Differentially expressed genes between offset time points during slender and stumpy differentiation. The offset comparison aligns slender 15 hrs with stumpy 0 hrs, where only 22 genes are differentially expressed. Genes with an absolute $\log_2FC > 2$ and an adjusted p -value < 0.01 were classified as differentially expressed. Corresponding volcano plots and detailed gene counts can be found in [Supplementary Figure 6](#).

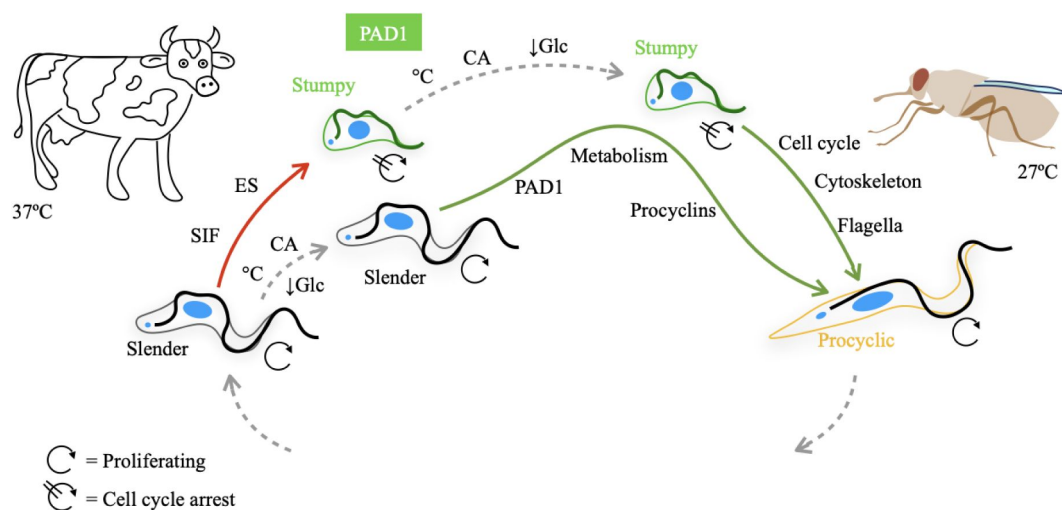


Figure 4

Slender and stumpy bloodstream forms must activate distinct pathways to transform into the procyclic form in the tsetse fly.

Differentiation to the PAD1-positive (green), cell cycle-arrested, short stumpy form can be triggered by either SIF or ES-attenuation (Zimmermann et al., 2017). Stumpy forms have a 2-3 day window to be ingested by a tsetse fly before they perish. When a tsetse fly takes a blood meal, it can ingest both slender and stumpy forms. Once in the fly's midgut, both forms begin their transformation into the procyclic fly form. During the initial 15 hrs, slender forms shift towards stumpy gene expression before diverging again. Stumpy forms need to reactivate the cell cycle, fully switch to proline metabolism, and elongate both their cytoskeleton and flagella (Supplementary Figure 7A). Slender forms must activate the essential PAD1 pathway, complete the switch to proline metabolism, and change to a procyclin coat (Supplementary Figure 7B). By 72 hrs into differentiation, both slender and stumpy forms have transitioned into the procyclic form. This Figure was adapted from Schuster et al. 2021.

Discussion

Historically, the lack of molecular markers to distinguish slender and stumpy forms resulted in conflicting reports on their transmissibility to the tsetse fly. Early experiments indicated that higher trypanosome levels in mammalian blood correlated with more infected flies. Since stumpy forms arise through a density-dependent mechanism, increased bloodstream concentrations inevitably meant more stumpy forms in mammalian hosts, an uncontrollable factor in these early studies (Robertson, 1912 [↗](#); Van Hoof, 1947 [↗](#); Baker & Robertson, 1957 [↗](#); Wijers, 1958 [↗](#)).

Moreover, the discovery of the secreted quorum sensing factor SIF in the 1990s reinforced the notion that this pathway serves to ensure the presence of stumpy forms (Reuner et al., 1997 [↗](#); Vassella et al., 1997 [↗](#)). It was also shown that stumpy forms begin developing traits needed for survival in the fly midgut, such as an elaborated mitochondrion and expressing enzymes associated with the Krebs cycle (Vickerman, 1965 [↗](#); Brown et al., 1973 [↗](#); Hamm et al., 1990 [↗](#); Reuner et al., 1997 [↗](#)). Previous research also suggested that the stumpy form was more adapted to life in the tsetse fly, pointing to their resistance to proteolytic stress and acidic conditions *in vitro* (Nolan et al., 2000 [↗](#)). Later, however, it was shown that the tsetse midgut is actually an alkaline environment, making the stumpy resistance to acidic conditions mute in the case of midgut survival (Liniger et al., 2003 [↗](#)).

Taken together, it is understandable why early studies concluded that only stumpy form trypanosomes could infect tsetse flies. This assumption, however, gave rise to the transmission paradox: under conditions of low parasitaemia during chronic infections, how can sufficient numbers of stumpy forms be maintained to ensure parasite transmission and sustain the disease cycle?

The ability to study the differences between slender and stumpy cells changed upon the discovery of the first stumpy specific molecular marker, the protein associated with differentiation 1 (PAD1), in 2009 (Dean et al., 2009 [↗](#)). Using the PAD1 marker, it was shown that skin-resident trypanosomes reveal a locally increased proportion of stumpy forms, providing a possible solution to this transmission paradox. In mice, the proportion of stumpy cells in the skin ranged from 8 - 80%, suggesting that these parasites might significantly contribute to infection dynamics (Capewell et al., 2016 [↗](#)).

Our recent study using artificial human skin models revealed that trypanosomes freshly deposited into the skin by tsetse flies develop into a quiescent form that cannot reinfect the vector again (Reuter et al., 2023 [↗](#)). Although this finding does not address parasites that migrate back into the skin from circulation, it highlights the complex dynamics occurring within tissue. Ultimately, the extent to which skin-resident populations contribute to tsetse transmission remains unclear, as direct quantification currently is not feasible.

In 2021, we offered an alternative explanation for the transmission paradox, namely that slender form trypanosomes can also infect and complete the life cycle in tsetse flies. All bloodstream form trypanosomes taken up in a bloodmeal have the potential to infect a mammalian host by establishing salivary gland infection in tsetse flies (Schuster et al., 2021 [↗](#)). While all our data supported this conclusion, some questions were raised, which we have now answered.

To mimic more natural conditions, we conducted infection experiments using slender forms without immune-suppressors, both male and female flies, and non-teneral flies.

The definition of a non-teneral and teneral fly is of great importance here.

The age of a teneral fly ranges anywhere from 8 to 72 hours post eclosion (hpe), with the important criterion that they have not taken up any blood (Otieno et al., 1983 [↗](#); Walshe et al., 2011 [↗](#)). Thus, when infecting teneral flies, their first bloodmeal is the infectious one. In contrast, we infected our non-teneral flies between 144 - 168 hpe, having received two regular bloodmeals, each two days apart, before the third infectious feeding.

While it could be argued that it would be better to use even older flies, the lifespan of the tsetse must be taken into consideration to ensure survival through the 30 days required for *T. brucei* to complete development in the vector. It has been suggested that tsetse flies in the wild may live longer than their laboratory raised counterparts, though estimated and observed life spans vary widely. The mean life span ranges from 35 to 178 days for female, and only 21 to 28 days for male flies (Jackson, 1949 [↗](#); Maudlin et al., 1999 [↗](#); Vale & Torr, 2005 [↗](#); Haines et al., 2020 [↗](#)). Regardless of the individual lifespan of a fly, a trypanosome infection is permanent.

We have demonstrated, in a statistically well-controlled manner, that a single slender trypanosome is capable of infecting the tsetse fly. Here, we unambiguously show that, in the absence of immunosuppressive treatment, slender forms can establish infections in tsetse flies, irrespective of the fly's age or sex.

The study by Ngoune et al. (2025) [↗](#) does not disprove this finding (Ngoune et al., 2025 [↗](#)). In their infection experiments, only 63% of the “stumpy” cells expressed PAD1, meaning that 37% must, by definition, have been slender forms. This heterogeneity complicates the interpretation of their results. It is also worth noting that Ngoune et al. used the AnTat 1.1E strain, which they maintained in culture without methylcellulose. We have observed that, during adaptation to methylcellulose-free medium, our *bona fide* pleomorphic AnTat 1.1 strain gradually loses its developmental competence. This could account for the heterogenous “stumpy” population reported by Ngoune et al.

Moreover, more than 40% of the adult male flies in their study did not survive the 28-day period required before dissection, further complicating the interpretation of their infection data.

In response to our original work Matthews & Larcombe postulated that “molecular characteristics” (PAD1 in particular) define a stumpy cell and that the eponymous morphological changes (e.g. stumpy formation) are no longer valid criteria (Matthews & Larcombe, 2022 [↗](#)). Although irreversible cell cycle arrest is still an accepted hallmark of stumpy trypanosomes, we cannot exclude that in the tsetse fly, the dividing slender population arrests the cycle and then transitions to the procyclic form. This scenario is of some theoretical interest, especially in view of the ongoing discussion about shallow vs. deep cell cycle arrest, e.g. in stem cell quiescence (Urbain & Cheung, 2021 [↗](#)). Therefore, we decided to conduct an RNA sequencing time course that would compare the transcriptional landscape of slender and stumpy bloodstream populations during differentiation to the procyclic form (**Figure 3** [↗](#)). Using fluorescent activated cell sorting, slender cultures were sorted to exclude any PAD1-positive cells, while stumpy cultures were sorted to include 100% PAD1 expressing cells.

At the outset, more than 300 genes were differentially expressed between the two bloodstream forms. However, after 72 hours (hrs) under differentiation conditions, no significant differences in gene expression remained; both slender and stumpy cells had transitioned into procyclic forms. Importantly, stumpy forms appear transcriptionally primed for rapid progression, while slender cells activate a distinct and temporally delayed gene expression program (**Supplementary Figure 8** [↗](#)).

The transient similarity observed at the specific timepoint, slender 15 hrs and stumpy 0 hrs, may reflect a brief convergence in gene expression, but is not indicative of a shared developmental pathway. Instead, the sustained differences in transcriptional profiles and enriched GO terms

(Supplementary Figures 5 [↗](#), 6 and 7) argue for fundamentally different regulatory mechanisms underlying the commitment to procyclic differentiation.

These data argue against a linear progression from slender to stumpy to procyclic. Instead, they suggest that slender forms transiently activate a subset of genes also expressed in stumpy cells but then follow a distinct transcriptional trajectory before ultimately converging with the stumpy pathway at 72 hrs, when both forms adopt a procyclic identity (Figure 3 [↗](#)). Thus, slender and stumpy forms remain transcriptionally distinct for at least the first 24 hrs of differentiation (Figure 3B [↗](#)). At this point, their transcriptomes do not correspond clearly to any of the three canonical forms - slender, stumpy, or procyclic. These findings demonstrate that slender forms can differentiate directly into procyclic forms without passing through a *bona fide* stumpy stage, i.e. without cell cycle arrest.

If slender forms can adopt a distinct transcriptional trajectory towards the procyclic state *in vitro*, there is no reason to assume that *T. brucei* could not use both routes - via stumpy or directly from slender - in the tsetse fly as well.

Additionally, there is one more observation that needs to be taken into account. We have shown that even small numbers of monomorphic trypanosomes strain 427 - incapable of differentiating into stumpy forms - can establish stable midgut infections in tsetse flies yet fail to progress to salivary gland colonization (Schuster et al., 2021 [↗](#)). These monomorphic cells do not respond to Stumpy Induction Factor (Reuner et al., 1997 [↗](#); Vassella et al., 1997 [↗](#)) and fail to upregulate PAD1 (Dean et al., 2009 [↗](#)), remaining locked in the proliferative slender bloodstream form. Thus, slender trypanosomes are indeed capable of infecting the tsetse midgut, even if they are monomorphic. The PAD pathway is not strictly required for colonization of the midgut. However, activation of the PAD pathway is essential for the generation of procyclic trypanosomes that are competent to complete development and colonize the salivary glands.

In retrospect, our results are perhaps less unexpected than initially assumed.

Material and Methods

Cell line

The pleomorphic *Trypanosoma brucei brucei* strain EATRO 1125 (serodome AnTat 1.1) (Le Ray et al., 1977) with an NLS-GFP PAD1 3'UTR molecular marker and an additional tdTomato fluorescence sequence (Reuter et al., 2023 [↗](#)) was used for infection experiments and cultured as previously described (Schuster et al., 2021 [↗](#)). For RNA sequencing experiments the same cell line was used with an additional EP1:YFP fusion protein (Schuster et al., 2021 [↗](#)).

Fly infection

Tsetse flies of the species *Glossina morsitans morsitans* were kept as previously described (Schuster et al., 2021 [↗](#)).

Teneral flies received their first, and infectious, bloodmeal 24-72 hours post-eclosion (hpe). They were infected with four trypanosomes per bloodmeal, either untreated or supplemented with 60 mM *N*-acetyl-glucosamine (NAG).

Non-teneral flies (144-168 hpe) were given two non-infectious bloodmeals, each two days apart, prior to the infectious bloodmeal containing 1×10^6 cells/ml of either slender or stumpy parasites. Prior to infection, fluorescence activated cell sorting was performed to ensure 100% slender or stumpy population.

Immunofluorescence

Cells were prepared, fixed and stained as previously described (Schuster et al., 2021 [DOI](#)).

Fluorescence activated cell sorting

Cell sorting was performed using the FACS Aria III (BD biosciences, Franklin Lakes, USA). Cells were harvested, resuspended in pre-warmed TDB at 1×10^7 cells/ml, transferred to FACS tubes through a 35 μ m cell strainer cap. Cells were first gated based on the tdTomato signal to exclude debris and residual medium. Subsequently, GFP fluorescence from the NLS-GFP PAD1 3'UTR reporter was analyzed. Depending on the desired population, gates were set to isolate either PAD1-positive (stumpy) or PAD1-negative (slender) cells with high purity. For each experiment, 1×10^6 cells were sorted using a 100 μ m nozzle at the lowest sorting speed to minimize mechanical stress. Sorted cells were collected into 15 ml tubes containing pre-warmed FCS, resulting in a final FCS concentration of 15% (v/v). After sorting, cell motility, concentration, and PAD1 signal were checked by microscopy.

Bulk RNA sequencing

Both slender and stumpy trypanosomes were differentiated into procyclic forms as previously described. The 0 hour (hr) timepoint was collected immediately before differentiation was induced, and additional samples were taken at 8, 15, 24, 72 hrs following addition of *cis*-aconitate.

Cells were harvested from biological triplicates, resuspended in 1 ml pre-warmed PBS containing 10 % FCS, and transported to the Helmholtz Institute for RNA Infection biology (HIRI) Würzburg in a 37°C incubation chamber.

Upon arrival, cells were washed twice with pre-warmed PBS to remove FCS in 1 ml of pre-warmed PBS 15 min prior to sorting, Calcein-AM violet was added at a final concentration of 1 μ M to label viable cells.

Live cells were sorted using a FACS Aria III cytometer (BD Biosciences). 1000 cells per replicate were deposited into single wells of a 48 well-plate containing 2.6 μ l of 1x lysis buffer (Takara) and 0.01 μ l of RNase inhibitor (40 U/ μ l; Takara). Sorting was performed in triplicates for each timepoint, and plates were immediately placed on ice and stored at -80°C.

Library preparation and sequencing were performed as previously described (Müller et al., 2018 [DOI](#)). Very briefly, lysates were supplemented with 0.2 μ l ERCC Spike-in Control Mix 1 (Thermo Fisher Scientific) at a 1:20,000,000 dilution. Libraries were prepared using the SMART-Seq v.4 Ultra Low Input RNA Kit (Takara), utilizing a quarter of the recommended reagent volumes. PCR amplification was performed for 27 cycles, and cDNA was purified with Agencourt AMPure XP beads (Beckman Coulter) and 15 μ l elution buffer (Takara). Library quantification was carried out using a Qubit 3 Fluorometer and dsDNA Hs Assay kit (Life Technologies), while quality assessment was performed by using a 2100 Bioanalyzer with High Sensitivity DNA kit (Agilent). 0.5 ng of cDNA was used as input for the Nextera XT (Illumina) tagmentation-based library preparation protocol. The reaction was performed at one quarter of the recommended volumes, with a 10-minute tagmentation step at 55 °C, and a 1-minute extension step during PCR. Libraries were pooled and sequenced in paired-end mode with 2x 75 cycles using Illumina's NextSeq 500.

Analysis of bulk RNA-sequencing data

Bulk RNA-sequencing analysis was performed according to scripts and online resources published by Berry et al., 2021 [DOI](#) using RStudio (version 4.02). Read demultiplexing and quality control were performed with FASTQC (version 0.12.1, (Andrews, 2012 [DOI](#))). Adaptors were removed and reads were trimmed using Fastp (version 0.23.4, (Chen, 2023 [DOI](#))).

The *T. b. brucei* TREU 972 reference genome was obtained from the Tritryp database (TritrypDB). rRNA was removed (Aslett et al., 2010 [↗](#)), and, to remove redundancy, only one representative from each gene group was retained while others were masked. Gene groups were defined based on sequence similarity using CD-Hit software (Li & Godzik, 2006 [↗](#)).

Kallisto was used for read alignment, and genes with ≥ 3 counts per million (cpm) in at least two samples were retained for downstream analysis. Data normalization was performed using the EdgeR package in RStudio (Robinson et al., 2009 [↗](#); Team R.C., 2021 [↗](#)).

SL152 (slender 15 hrs, 2nd replicate) and SL241 (24 hrs, 1st replicate) were considered outliers based on hierarchical clustering distance using the Minkowski metric (Lee & Willcox, 2014 [↗](#)). Differentially expressed genes (DEGs) were identified using DESeq2, EdgeR, and Limma packages in RStudio (Love et al., 2014 [↗](#); Ritchie et al., 2015 [↗](#); Robinson et al., 2009 [↗](#); Team R.C., 2020), applying significance criteria of log2 fold change > 1 and p value < 0.01 .

Statistics

Two-tailed Fisher's exact tests were performed for all fly infection data using GraphPad prism version 9.0.0 for Windows (Graphpad software, San Diego, California USA, <http://www.graphpad.com> [↗](#)).

No. of flies infected	Fly sex	No. of slender cells/ml	Concentration of NAG	Days until dissection	No. of flies dissected	MG	PV	SG
37	male	200	0 mM	35	16	0	0	0
19	female	200	0 mM	35	17	1	1	0
24	male	200	0 mM	35	24	5	4	1
24	female	200	0 mM	35	22	2	2	0
24	male	200	0 mM	35	21	1	1	0
9	female	200	0 mM	35	9	1	1	0

No. of flies infected	Fly sex	No. of slender cells/ml	Concentration of NAG	Days until dissection	No. of flies dissected	MG	PV	SG
37	male	200	60 mM	35	35	3	3	2
19	female	200	60 mM	35	18	1	1	0
26	male	200	60 mM	35	24	1	1	1
27	female	200	60 mM	35	27	1	1	0
26	male	200	60 mM	35	26	7	4	2
9	female	200	60 mM	35	8	3	3	0

Supplementary Figure 1

Absolute numbers of tsetse fly infections using slender bloodstream forms of *T. brucei*, with or without the addition of *N*-Acetyl-Glucosamine (NAG).

Both, male and female flies were infected with blood containing 200 slender cells per ml of blood, either untreated or supplemented with the immune-suppressing chemical, *N*-Acetyl-Glucosamine (NAG, 60mM). Tsetse flies have an estimated drinking volume of 20 μ l (Gibson & Bailey, 2003), which results in an uptake of 4 parasites per bloodmeal. All flies were dissected 35 days post infection, and their midgut (MG), proventriculus (PV), and salivary glands (SG) examined for parasite presence. Prior to infection, slender cells of the tdTomato NLS-GFP:PAD1 3'UTR line were verified to lack *pad1* expression, confirming pure slender identity (< 0.05% PAD1 positive).

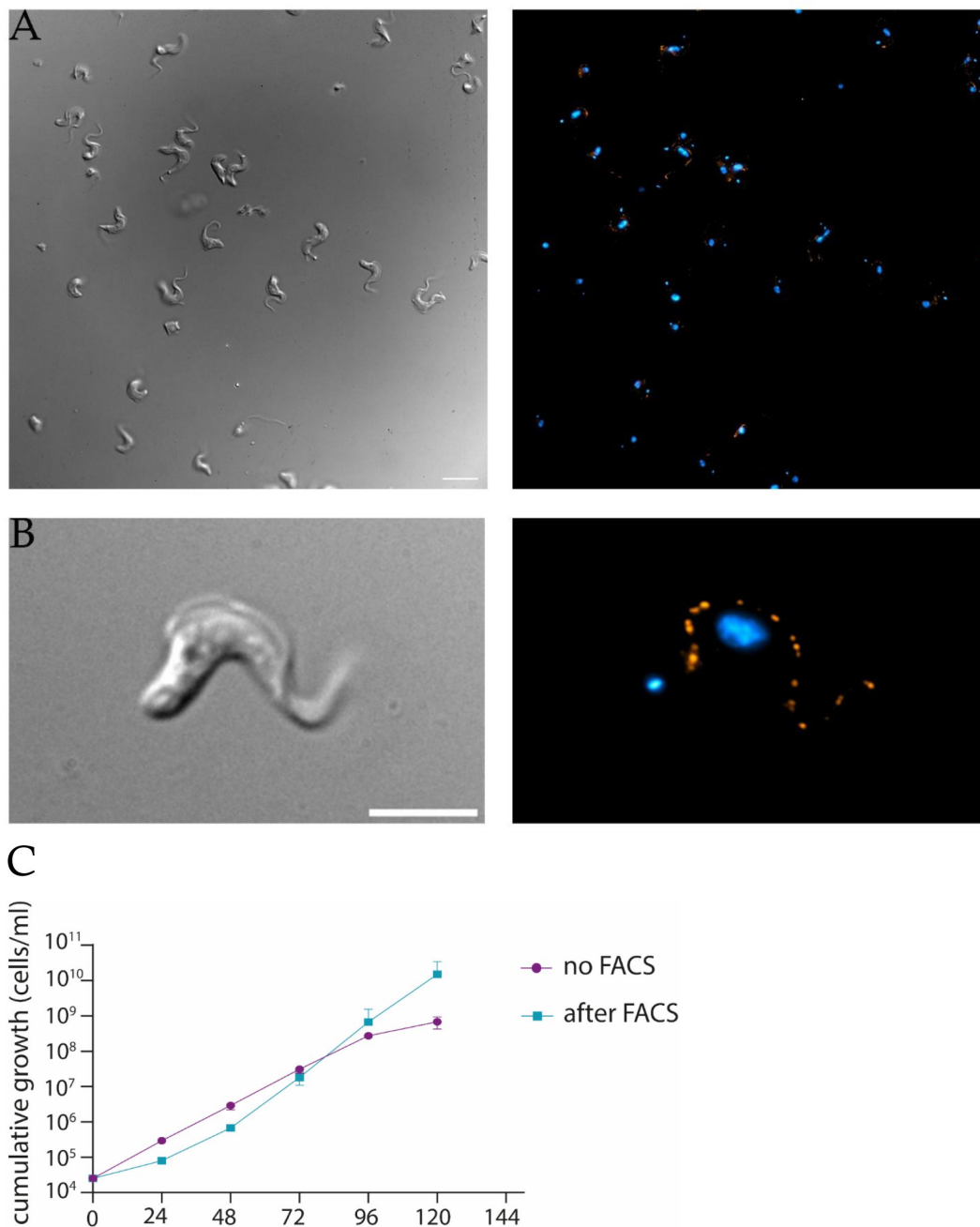
No. of flies infected	Fly sex	No. of slender cells/ml	Concentration of NAG	Days until dissection	infectious feed	No. of flies dissected	MG	PV	SG
20	male	1x10 ⁶	0 mM	30	3 rd	17	0	0	0
20	male	1x10 ⁶	0 mM	30	3 rd	19	2	2	2
21	male	1x10 ⁶	0 mM	35	3 rd	21	5	4	2
20	male	1x10 ⁶	0 mM	35	3 rd	19	0	0	0
20	female	1x10 ⁶	0 mM	30	3 rd	20	1	0	0
21	female	1x10 ⁶	0 mM	30	3 rd	21	2	2	2
20	female	1x10 ⁶	0 mM	35	3 rd	20	0	0	0
20	female	1x10 ⁶	0 mM	35	3 rd	20	0	0	0

No. of flies infected	Fly sex	No. of stumpy cells/ml	Concentration of NAG	Days until dissection	infectious feed	No. of flies dissected	MG	PV	SG
20	male	1x10 ⁶	0 mM	38	3 rd	20	2	2	1
21	male	1x10 ⁶	0 mM	36	3 rd	16	1	1	1
23	male	1x10 ⁶	0 mM	35	3 rd	19	2	2	0
14	male	1x10 ⁶	0 mM	35	3 rd	11	0	0	0
20	female	1x10 ⁶	0 mM	38	3 rd	19	1	0	0
20	female	1x10 ⁶	0 mM	36	3 rd	20	1	0	0
23	female	1x10 ⁶	0 mM	35	3 rd	19	0	0	0
15	female	1x10 ⁶	0 mM	35	3 rd	13	2	1	0

Supplementary Figure 2

Absolute numbers of infections in non-teneral tsetse flies with either slender or stumpy *T. brucei* cells.

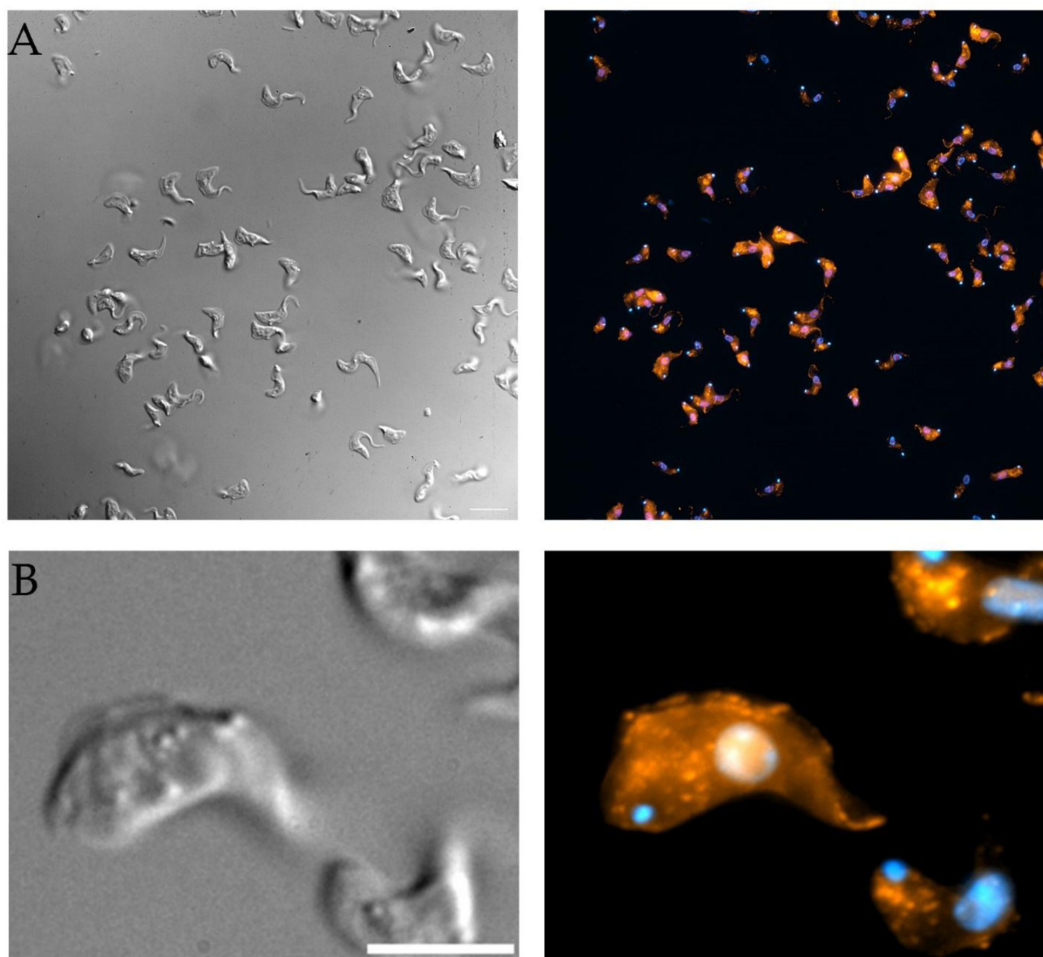
Male and female flies were infected with untreated blood containing 1x10⁶ cells/ml. The tdTomato NLS-GFP:PAD1 3'UTR cell line enabled FACS-based separation of stumpy (PAD1 positive, GFP in nucleus) and slender (PAD1 negative, no nuclear fluorescence) forms prior to infection. All non-teneral flies were 144-168 hours post eclosion (hpe) and had already received two non-infectious bloodmeals prior to the infectious feed. Flies were dissected 35 days post infection, and their midgut (MG), proventriculus (PV), and salivary glands (SG) examined for parasite presence.



Supplementary Figure 3

Slender *T. brucei* cells of the tdTomato NLS-GFP:PAD1 3'UTR line do not express PAD1 or exhibit stress responses following fluorescence activated cell sorting (FACS).

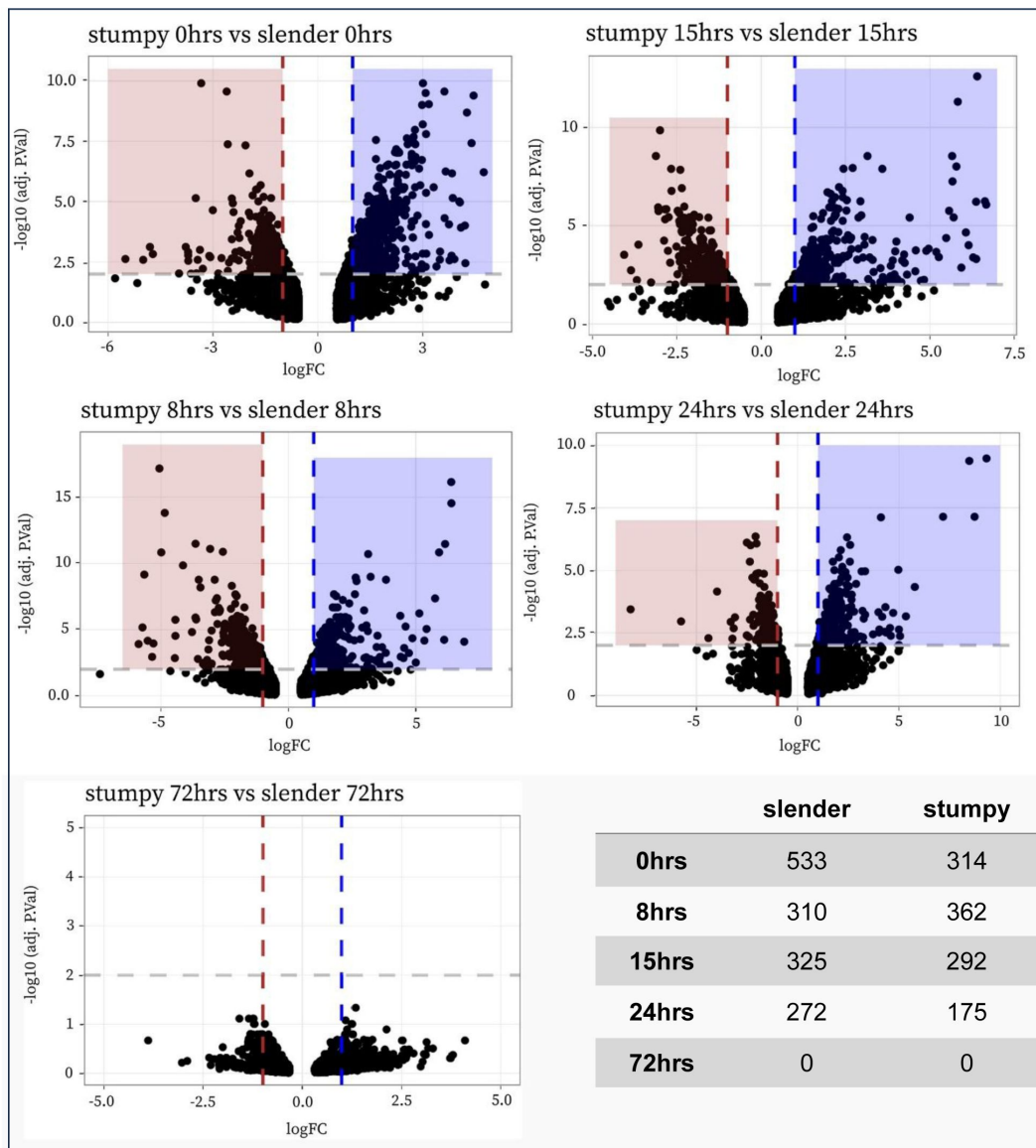
FACS was used to ensure pure slender populations (*pad1* negative) prior to infection. A: Immunofluorescence (IF) images of slender cells immediately after FACS, confirming sorting success. Parasites were fixed in 4% Paraformaldehyde (PFA), stained with DAPI (blue), and labelled with an anti-PAD1 antibody (orange); scalebar: 20 μ m. B: High-resolution IF image of a single slender trypanosome post sorting, showing clear absence of nuclear *pad1* signal; scalebar: 10 μ m. C: Growth curves comparing slender cells post sorting (green) and untreated slender cells (purple), indicating no growth impairment due to sorting.



Supplementary Figure 4

Fluorescence activated cell sorting (FACS) was used to isolate stumpy cells of the tdTomato NLS-GFP:PAD1 3'UTR line, ensuring a pure stumpy population prior to infection.

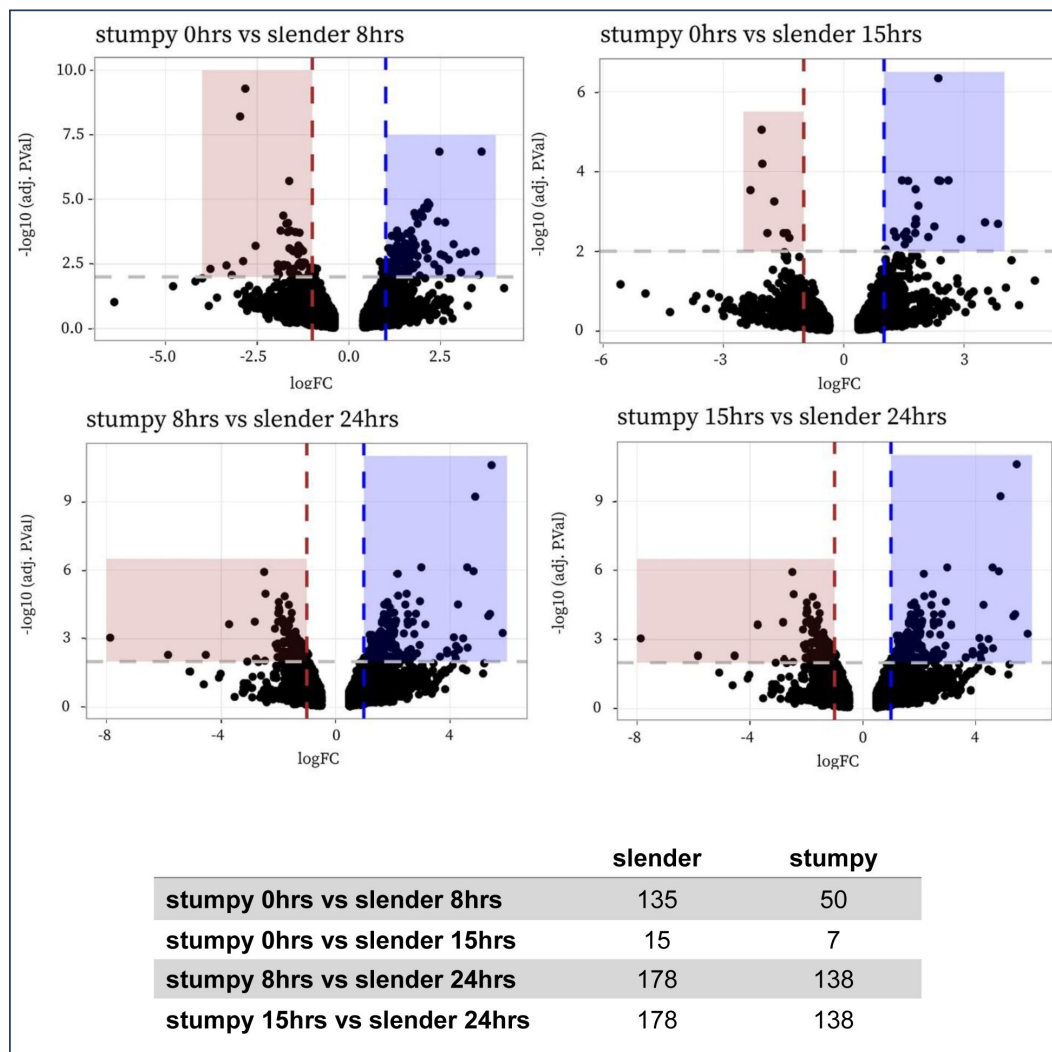
A: Stumpy cells from a SIF-induced stumpy culture (grown to 5×10^5 cells/ml and kept for 48 hours), after FACS to confirm sorting success. Cells were fixed in 4% PFA immediately after sorting, stained with DAPI (blue), and labelled with an anti-PAD1 antibody (orange); scalebar: 20 μ m. B: High-resolution IF image of a single stumpy trypanosome displaying characteristic stumpy morphology and strong *pad1* signal (orange); scalebar: 10 μ m.



Supplementary Figure 5

Volcano plots showing differential gene expression between stumpy (red) and slender (blue) *T. brucei* forms during *in vitro* differentiation to the procyclic form at 0, 8, 15, 24 and 72 hrs after induction.

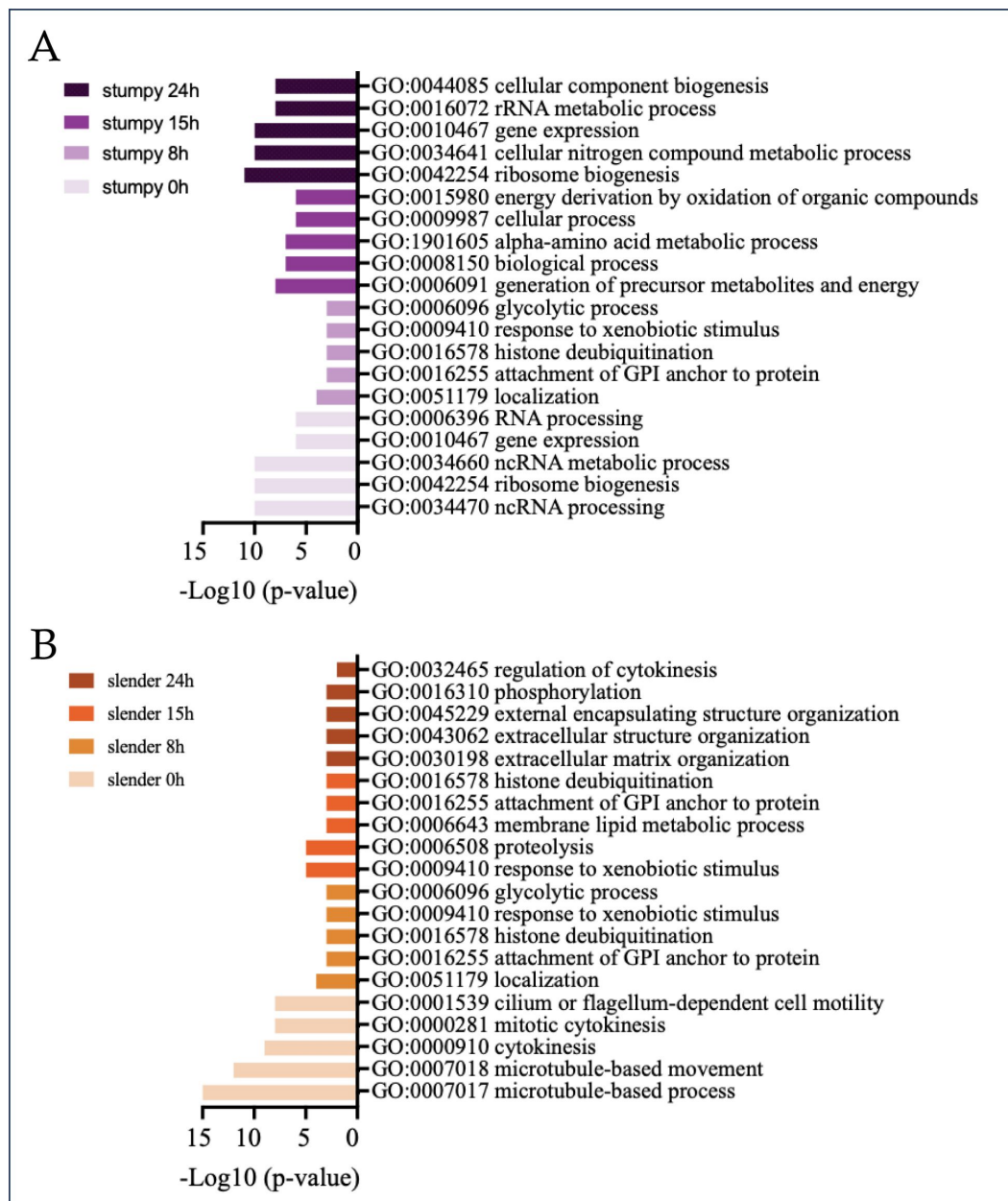
Differentiation was initiated by adding *cis*-aconitate, lowering the temperature to 27°C, and depleting glucose in the medium. Each black dot represents one gene. Genes within the coloured boxes show 2x up-regulation in slender (blue dotted line) or 2x up-regulation in stumpy (red dotted line) cells, with a p-value ≤ 0.01 (grey dotted line). Red and blue boxes highlight significantly upregulated genes in stumpy or slender, respectively, with the exact gene counts listed in the accompanying table. $\log_{2} \text{FC}$ = $\log_{2}$ Fold Change; hrs = hours after the addition of *cis*-aconitate.



Supplementary Figure 6

Volcano plots showing differential gene expression of stumpy (red) and slender (blue) *T. brucei* forms following *in vitro* differentiation to the procyclic form.

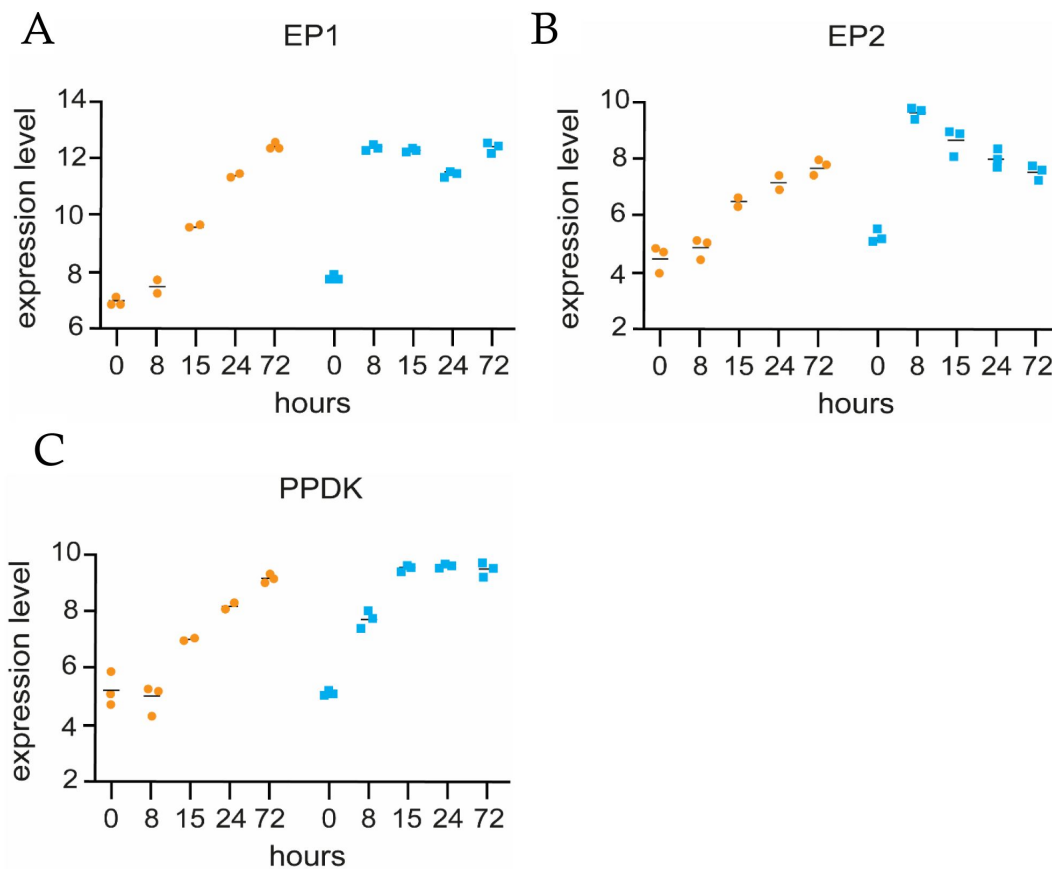
Differentiation was initiated by adding *cis*-aconitate, lowering the temperature to 27°C, and depleting glucose. An offset comparison – based on proximity in the PCA plot ([Figure 3A](#)) – aligns slender cells at 15 hours (hrs) with stumpy cells at 0 hrs. Each black dot represents one gene. Genes within the coloured boxes show 2x up-regulation for slender (blue dotted line) or 2x up-regulation for stumpy (red dotted line) cells, with a p-value ≤ 0.01 (grey dotted line). Red and blue boxes highlight significantly upregulated genes in stumpy or slender, respectively, with the exact numbers of differentially expressed genes listed in the accompanying table. Notably, slender cells at 15 hrs exhibit a similar gene expression profile to that of stumpy trypanosomes after 0 hrs, before diverging again at later time points. logFC= log2 Fold Change; hrs = hours after the addition of *cis*-aconitate.



Supplementary Figure 7

During differentiation into the procyclic form, slender and stumpy parasites exhibit gene expression profiles associated with distinct biological processes and molecular functions.

Gene Ontology (GO) enrichment analysis between corresponding time points identified genes with a log₂ fold change of at least 1 (indicating 2x expression) and a p-value of ≤ 0.01 for either slender (orange) or stumpy (purple) forms at 0, 8, 15, 24, and 72 hours (Supplementary Figure 5). GO annotations were sourced from the *TriTryp.org* [TriTryp.org](https://trityp.org) database ([TriTryp.org](https://trityp.org)) and refined using Revigo. The most significantly enriched GO terms for each time point are shown.



Supplementary Figure 8

Dot plots showing expression levels of genes associated with the procyclic form - EP1 (A), EP2 (B) and pyruvate phosphate dikinase (PPDK) (C) - in slender (orange) and stumpy (blue) forms during *in vitro* differentiation.

Differentiation was induced by the addition of *cis*-aconitate, reduction of temperature to 27°C, and glucose depletion. Stumpy forms reach procyclic-like expression levels by approximately 8 hours, whereas slender forms display a more gradual increase. RNA-sequencing was performed in triplicates; each coloured dot represents one replicate (1000 cells) and the black line indicates the mean expression value.

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Reviewer #1 (Public review):

Summary:

This work provides evidence that slender *T. brucei* can initiate and complete cyclical development in *Glossina morsitans* without GlcNAc supplementation, in both sexes, and importantly in non-teneral flies, including salivary-gland infections.

Comparative transcriptomics show early divergence between slender- and stumpy-initiated differentiation (distinct GO enrichments), with convergence by ~72 h, supporting an alternative pathway into the procyclic differentiation program.

The work addresses key methodological criticisms of earlier studies and supports the hypothesis that slender forms may contribute to transmission at low parasitaemia.

Strengths:

- (1) Directly tackles prior concerns (no GlcNAc, both sexes, non-teneral flies) with positive infections through to the salivary glands.
- (2) Transcriptomic time course adds some mechanistic depth.
- (3) Clear relevance to the "transmission paradox"; advances an important debate in the field.

Weaknesses:

- (1) Discrepancy with Ngoune et al. (2025) remains unresolved; no head-to-head control for colony/blood source or microbiome differences that could influence vector competence.
- (2) Lacks in vivo feeding validation (e.g., infecting flies directly on parasitaemic mice) to strengthen ecological relevance.
- (3) Mechanistic inferences are largely correlative (although not requested, there is no functional validation of genes or pathways emerging from the transcriptomics).
- (4) Reliance on a single parasite clone (AnTat 1.1) and one vector species limits external validity.

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Reviewer #2 (Public review):

Summary:

This paper is an exciting follow-up to two recent publications in eLife: one from the same lab, reporting that slender forms can successfully infect tsetse flies (Schuster, S et al., 2021), and another independent study claiming the opposite (Ngoune, TMJ et al., 2025). Here, the authors address four criticisms raised against their original work: the influence of N-acetylglucosamine (NAG), the use of teneral and male flies, and whether slender forms bypass the stumpy stage before becoming procyclic forms.

Strengths:

We applaud the authors' efforts in undertaking these experiments and contributing to a better understanding of the *T. brucei* life cycle. The paper is well-written and the figures are clear.

Weaknesses:

We identified several major points that deserve attention.

- (1) What is a slender form? Slender-to-stumpy differentiation is a multi-step process, and most of these steps unfortunately lack molecular markers (Larcombe et al, 2023). In this paper, it is essential that the authors explicitly define slender forms. Which parameters were used? It is implicit that slender forms are replicative and GFP::PAD1-negative. Isn't it possible that some GFP::PAD1-negative cells were already transitioning toward stumpy forms, but not yet expressing the reporter? Transcriptomically, these would be early transitional cells that, upon exposure to "tsetse conditions" (in vitro or in vivo), could differentiate into PCF through an alternative pathway, potentially bypassing the stumpy stage (as suggested in Figure 4).

Given the limited knowledge of early molecular signatures of differentiation, we cannot exclude the possibility that the slender forms used here included early differentiating cells. We suggest:

1.1 Testing the commitment of slender forms (e.g., using the plating assay in Larcombe et al., 2023), assessing cell-cycle profile, and other parameters that define slender forms.

1.2 In the Discussion, acknowledging the uncertainty of "what is a slender?" and being explicit about the parameters and assumptions.

1.3 Clarifying in the Materials and Methods how cultures were maintained in the 3-4 days prior to tsetse infections, including daily cell densities. Ideally, provide information on GFP expression, cell cycle, and morphology. While this will not fully resolve the concern, it will allow future reinterpretation of the data when early molecular events are better understood.

(2) Figure 1: This analysis lacks a positive control to confirm that NAG is working as expected. It would strengthen the paper if the authors showed that NAG improves stumpy infection. Once confirmed, the authors could discuss possible differences in the tsetse immune response to slender vs. stumpy forms to explain the absence of an effect on slender infections.

(3) Figure 2. To conclude that teneral flies are less infected than non-teneral flies, data from Figures 1 and 2 must be directly comparable. Were these experiments performed simultaneously? Please clarify in the figure legends. Moreover, the non-teneral flies here are still relatively young (6-7 days old), limiting comparisons with Ngoune, TMJ et al. 2025, where flies were 2-3 weeks old.

(4) Figure 3. The PCA plot (A) appears to suggest the opposite of the authors' interpretation: slender differentiation seems to proceed through a transcriptome closer to stumpy profiles. Plotting DEG numbers (panel C) is informative, but how were paired conditions selected? Besides, plotting of the number of DEGs between consecutive time points within and between parasite types is also necessary. There may also be better computational tools to assess temporal relationships. Finally, how does PAD1 transcript abundance change over time in both populations? It would also be important to depict the upregulation of procyclic-specific genes.

(5) Could methylcellulose in the medium sensitize parasites to QS-signal, leading to more frequent and/or earlier differentiation, despite low densities? If so, cultures with vs. without methylcellulose might yield different proportions of early-differentiating (yet GFP-negative) parasites. This could explain discrepancies between the Engstler and Rotureau labs despite using the same strain. The field would benefit from reciprocal testing of culture conditions. Alternatively, the authors could compare infectivity and transcriptomes of their slender forms under three conditions: (i) in vitro with methylcellulose, (ii) in vitro without methylcellulose, and (iii) directly from mouse blood.

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